Portal hypertension

Dr. Elham Ahmed Hassan

Assistant Professor of Gastroenterology and Tropical Disease

Definition

- Portal hypertension is defined as persistent portal venous pressure above normal range (7-10 mmHg).

- Portal hypertension is directly responsible for the two major complications of cirrhosis:
  - variceal hemorrhage
  - ascites.

- Variceal hemorrhage is an immediate life-threatening with 20–30% mortality rate associated with each episode of bleeding

Pathophysiology

- **Portal hypertension results from:**

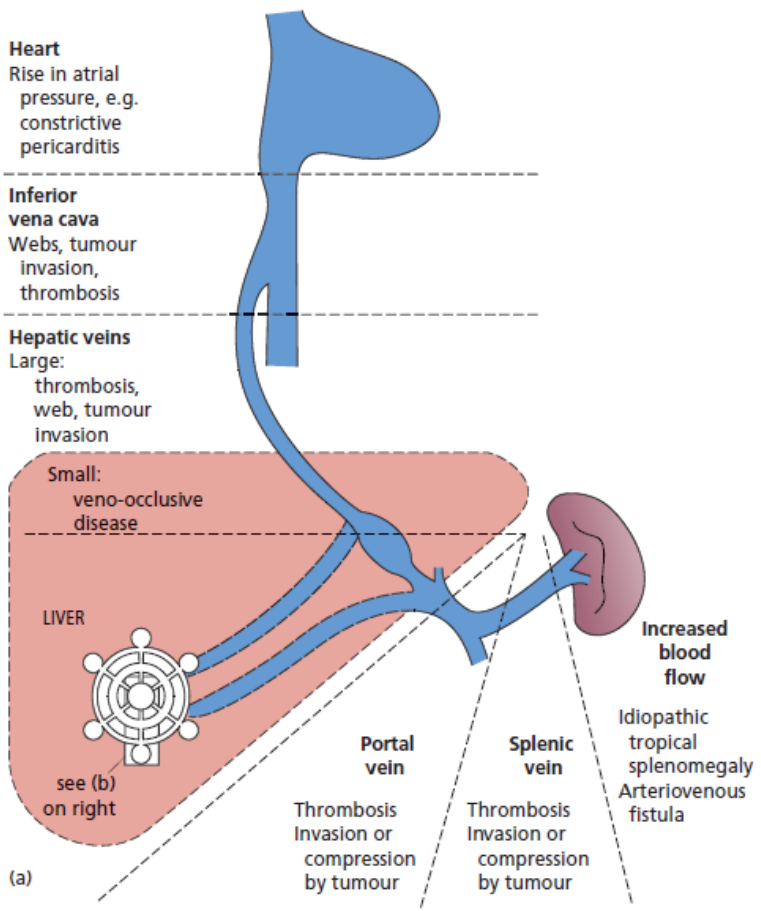
  - Increase in portal resistance

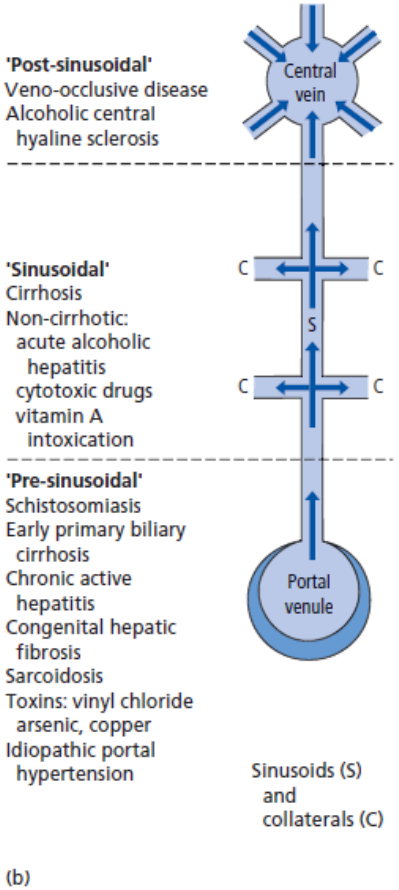
  - Increase in portal inflow.

Causes

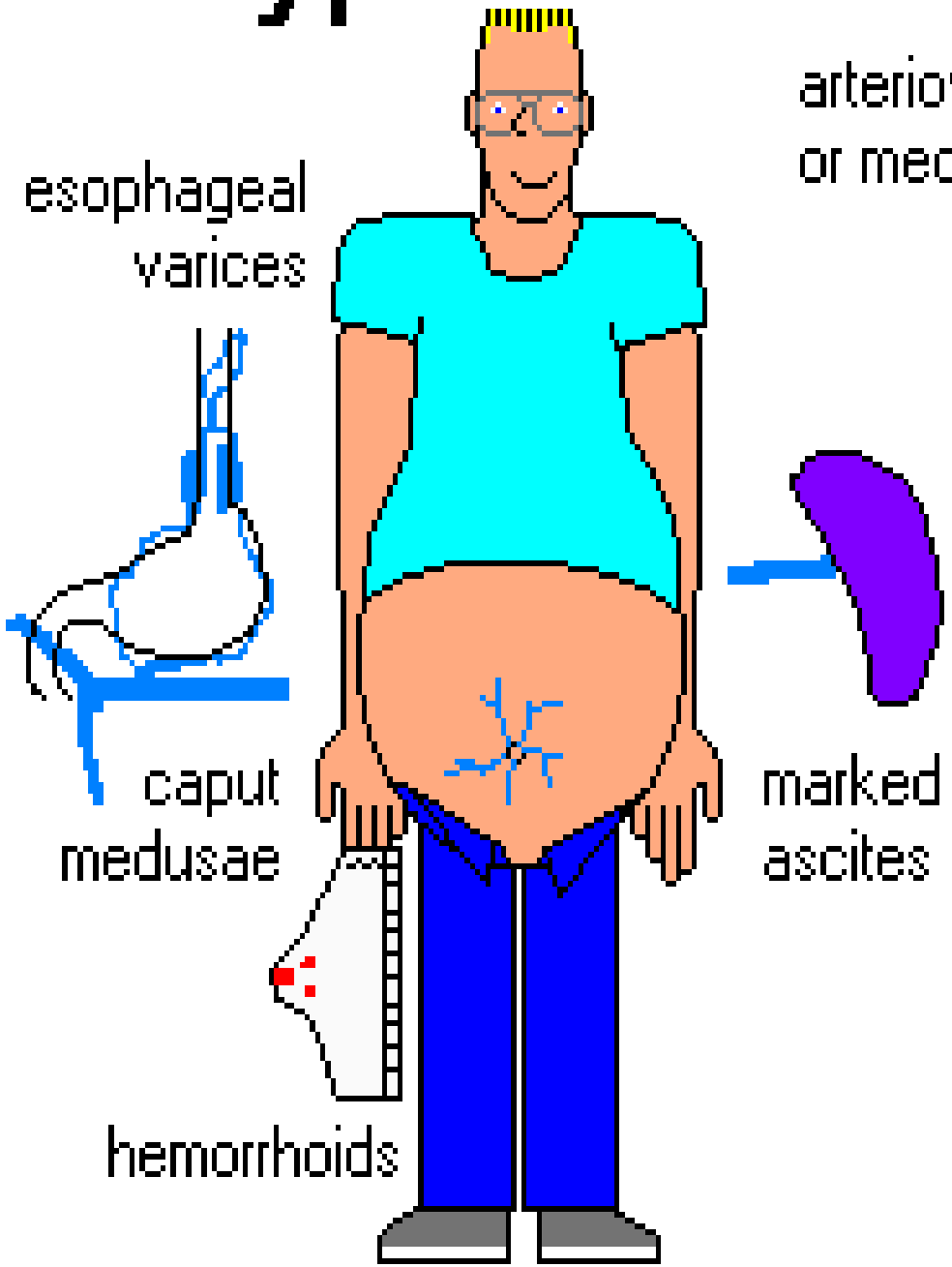
- **Pre-hepatic type:** Stenosis, thrombosis, infection, Trauma
- **Intra-hepatic type:** liver cirrhosis
- **Post-hepatic type:** CHF, IVC thrombosis

Causes of portal hypertension





Portal Hypertension



arteriovenous shunting and/
or mechanical obstruction

esophageal
varices

hypersplenism:
-- moderate anemia
-- neutropenia
-- thrombocytopenia

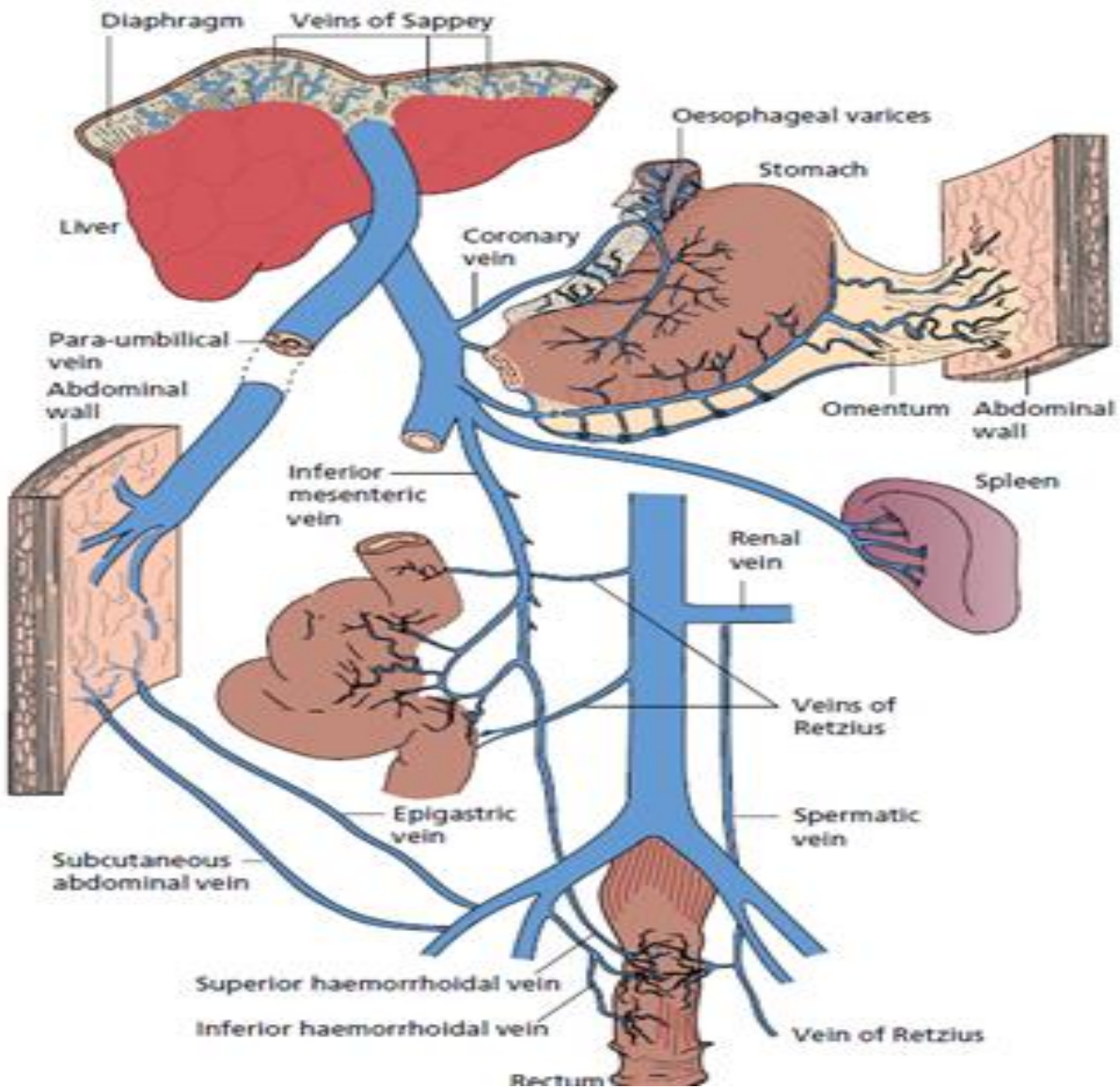
caput
medusae

marked
ascites

hemorrhoids

Of course,
liver failure and
its signs are often
present as well.

Sites of portal-systemic collateral circulation in liver cirrhosis



Investigations

A- Upper Gastrointestinal endoscopy

- the most commonly used method to detect varices.

- all patients with cirrhosis screened for esophageal varices.

- no varices on initial endoscopy, endoscopy repeated in 2 to 3 years.

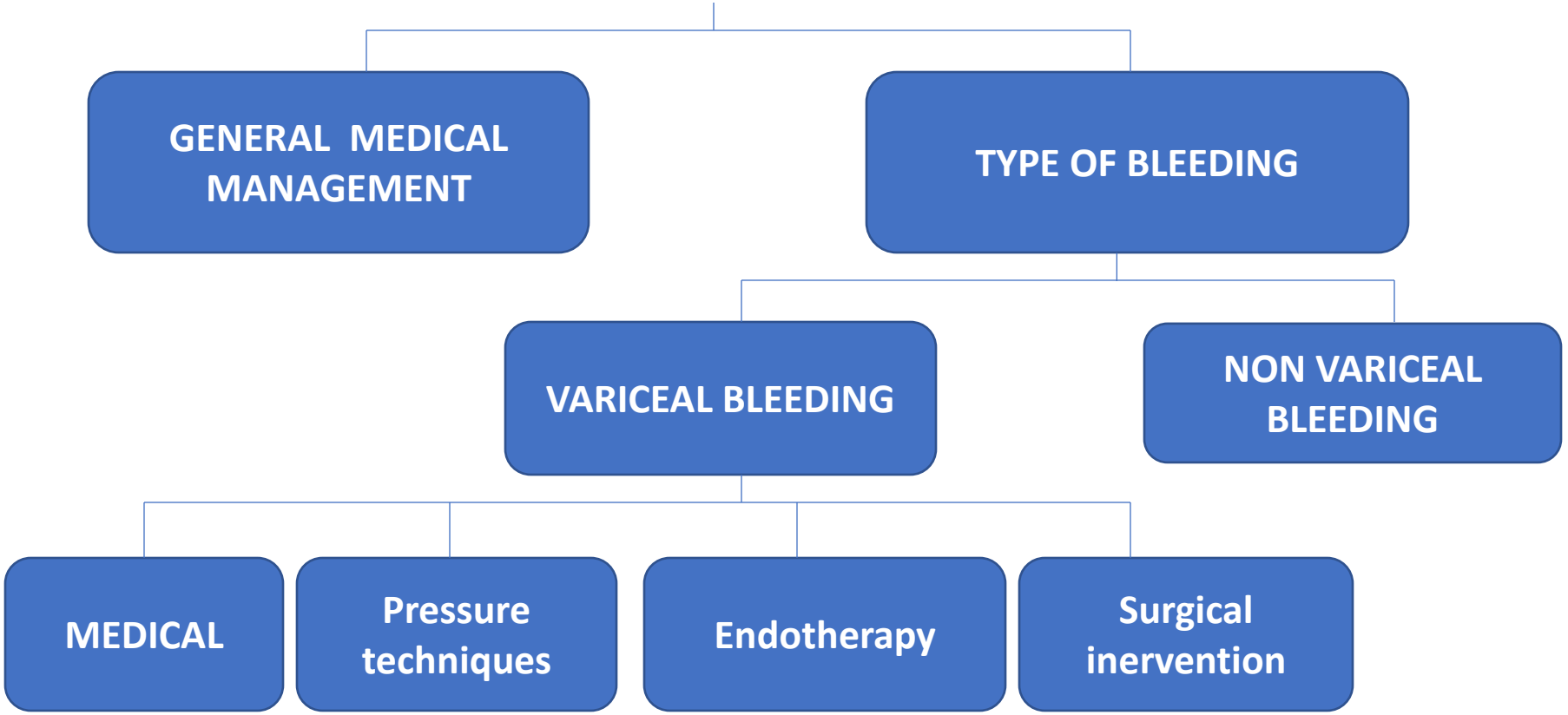
- small varices on the initial endoscopy, endoscopy repeated in 1 to 2 years.

B- Ultrasound (Dopplar)

C- Endoscopic ultrasound

D- Others e.g. non-invasive tests

Management of UGIB



A. Medical Therapy (2 groups)

(a) Drugs decrease splanchnic blood flow (VC)

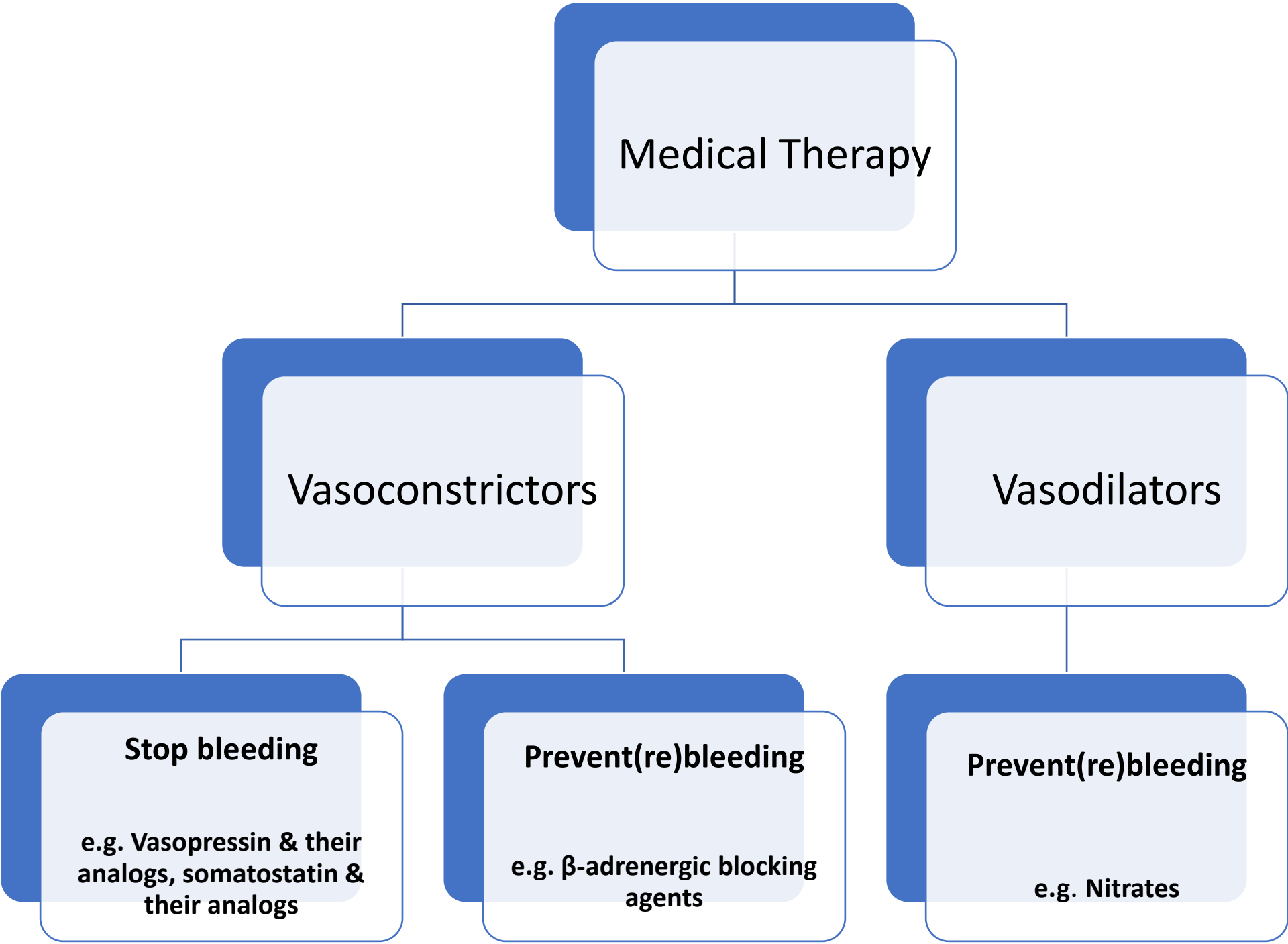
- Stop bleeding e.g. Vasopressin & their analogs, somatostatin & their analogs,

- Prevent (re)bleeding e.g. β-adrenergic blocking agents e.g. propranolol

(b) Drugs decrease intrahepatic vascular resistance (VD)

e.g. α-adrenergic blocking agents (prazosin), angiotensin receptor blocking agents and nitrates

(but only nitrates are now considered for clinical use).


Medical Therapy
Vasoconstrictors
Vasodilators
Stop bleeding
e.g. Vasopressin & their analogs, somatostatin & their analogs
Prevent(re)bleeding
e.g. β-adrenergic blocking agents
Prevent(re)bleeding
e.g. Nitrates

Vasoconstrictors

Vasopressin and Its Analogs

1- Vasopressin (Pitressin): (0.1–0.5 units/min for 4 to 12hrs (up to 48hrs)

- is an endogenous peptide hormone that causes splanchnic V.C, reduces portal venous inflow, reduces portal pressure.

- It is associated with serious systemic side effects:

  * necrosis of the bowel.

  * direct negative inotropic and chronotropic effects on myocardium that lead to reduced cardiac output and bradycardia,

  * antidiuresis can result in hyponatremia.

  * coronary artery constriction

- Vasopressin is rarely used.



2- Terlipressin (Glypressin)(2 mg bolus followed by 1mg/4-6 hr for 48hrs,1mg /4hr for 3days)

- A synthetic analogue of vasopressin (cleaved by endothelial peptidases to release lysine vasopressin).

- Compared with vasopressin, terlipressin has

  - lower systemic side effects (↑BP, coronary vasospasm) .
  - longer biologic activity
  - preserving renal function, particularly important feature in cirrhotic pts.
  - associated with improved survival.

- As adjuvant to endoscopic ttt.
- Other uses: norepinephrine-resistant septic shock and hepatorenal syndrome


Caja con 1 frasco ámpula con liofilizado y
1 ampolleta con 5 ml de diluyente
Glypressin®
Terlipresina
Solución Inyectable
FORMULA:
El frasco ámpula con liofilizado contiene:
Acetato de Terlipresina ..... 1 mg
(Equivalente a 0.86 mg de Terlipresina)
Excipiente cs
La ampolleta con diluyente contiene:
Cloruro de sodio ..... 45 mg
Agua inyectable csp ... 5 ml
FERRING
Glypressin
Terlipresina
FERRING


Terlipressin Injection
1 mg/10 ml Sterile Solution for IV use
10 ml Ampoule
Terlimed
Medwise
Terlimed


Rx Terlipressin Injection
TERLISIL™
Sterile Solution for IV Use Only
TERLISIL

⦿ Somatostatin and Its Analogues

1- Somatostatin

- is a 14-amino-acid peptide.

- vasoconstriction, somatostatin diminishes blood flow to the portal system, thus decreasing variceal bleeding.

- has effects similar to those of vasopressin but does not cause coronary vasoconstriction

- Half-life: Following IV injection, 1 - 3 minutes; therefore, longer-acting analogs have been synthesized.

2- Octreotide (T1/2 80-120 min), lanreotide, and vapreotide.



- Octreotide (Sandostatin) is a synthetic analogue of somatostatin

- Action: It decreases portal pressure and collateral blood flow by inhibiting release of glucagon involved in vasodilatation

- Compared with somatostatin, octreotide has similar pharmacologic actions with greater potency and longer duration of action.

- Initial bolus of 50 µg followed by a continuous infusion of 50 µg/h

- Mostly treatment with somatostatin or octreotide is combined with endoscopic management of variceal bleeding.

Non-selective β- Adrenergic Blocking Agents:

Not be used in bleeding attack

- **prevent variceal (re)bleeding** by reducing portal pressure by reducing portal and collateral blood flow.
- Monitoring efficacy by observing a decrease in HR:

  Dose increased gradually/3-5 days till HR of 25% below baseline or 55-60 b/min or maximum tolerated dose is reached, provided Syst. BP >90mmHg.

- **E.g. propranolol** or **nadolol, timolol**
- Usual dose of propranolol is 40mg once daily
- side effects overemphasized as 15% need to discontinue the drug
- In intolerant or contraindications to beta blockers, **isosorbide mononitrate** has been tried.

Vasodilators

(not used in bleeding attack)

Nitrates

- decrease portal flow by decreasing mean arterial pressure (by relaxation of vascular smooth muscle by stimulating intracellular cyclic guanosine monophosphate production).
- Decrease in cardiac output, rise in splanchnic arterial vascular resistance and decrease in porto-hepatic resistance
- Nitrates are no longer recommended, either alone or with a beta blocker, for primary prophylaxis
- For secondary prophylaxis, isosorbide mononitrate may be added to a beta blocker.

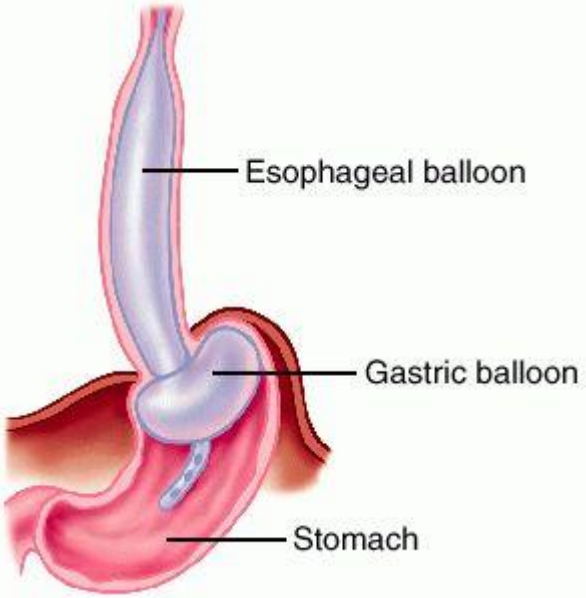


Antibiotic prophylaxis

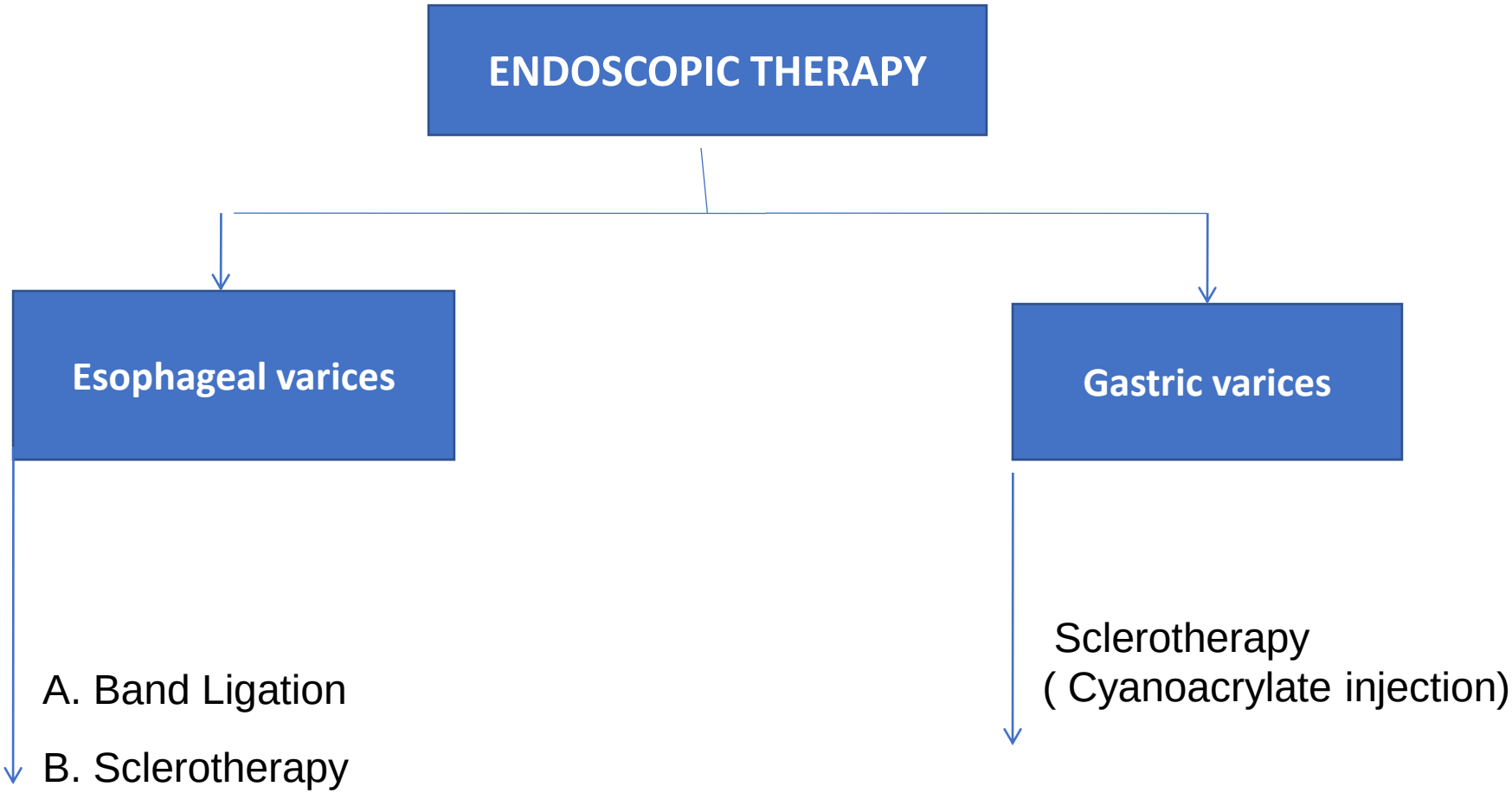
- As high risk of bacterial infections, which relates to early rebleeding rates and to a high mortality
- The recommended antibiotic is **norfloxacin**, in a dose of 400 mg orally twice a day, or **ciprofloxacine** 200 mg IV twice a day if the oral route is not possible.
- In patients with advanced cirrhosis : "Child–Pugh B or C", **ceftriaxone** proved to be more effective than oral norfloxacine

B. Pressure techniques

e.g. Sengstaken blakemore tube,



- Balloon should be inflated for less than 24 hrs.
- Initial control of bleeding in 85% to 98% of cases.
- Up to 75% rebleeding rate after balloon deflation.
- Serious complications, such as aspiration pneumonia, esophageal ulcer, rupture, and airway obstruction.


ENDOSCOPIC THERAPY
Esophageal varices
Gastric varices
A. Band Ligation
B. Sclerotherapy
Sclerotherapy
(Cyanoacrylate injection)

Esophageal varices

(a) Sclerotherapy

* has largely been supplanted by endoscopic band ligation,

* The sclerosants used :

- ethanolamine oleate (Ethamolin)
- sodium tetradecyl sulfate, (Sotradecol).
- sodium morrhuate, (Scleromate)
- Polidocanol(3%) {lauromacrogol}
- absolute alcohol

* the choice of a sclerosant based on availability, rather than efficacy.

(b) Endoscopic variceal ligation

* is the preferred endoscopic modality

* is simpler to perform than sclerotherapy.

* is fewer complications than sclerotherapy.

Gastric varices

- 25% of patients with portal hypertension have gastric varices,

- endoscopic therapy for fundal gastric variceal bleeding is injection of polymers of cyanoacrylate.

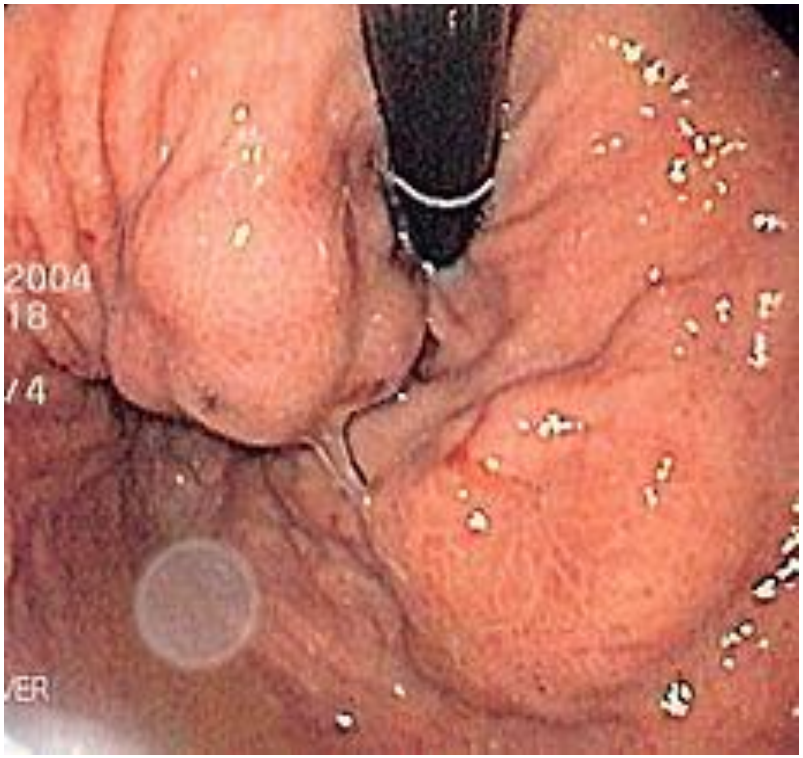
Esophageal varices





Gastric varices





C) Other therapy

- TIPS (Transjugular intrahepatic portosystemic shunt)
- covered esophageal stent
- Surgical treatment

- Esophageal Transection

- Portosystemic shunt procedures

- liver transplantation

THANK YOU